TRADING 212 QUANTITATIVE ALGORITHMIC TRADING BOT

High-Beta Momentum Screener, Compounding Position Sizing & Proxmox LXC Architecture

PLATFORM: TRADING 212 API v0 | UNIVERSE: 25 HIGH-BETA MOVERS | TIMEFRAME: 15-MIN | SLOTS: 3 CONCENTRATED (33% CAP)

1. EXECUTIVE OVERVIEW & MODULAR SYSTEM ARCHITECTURE

Engineered as a short-term momentum compounding machine, the bot scans 25 high-beta US growth equities every 15 minutes, ranks them 0-100 across four quantitative pillars, and concentrates capital into the top 3 highest-conviction breakouts.

1. DataHandler	2. StockScreener	3. Gemini AI Mgr	4. RiskManager	5. OrderExecutor
OHLCV & L1 Quotes	**25 High-Beta Stocks**	**News & Slot Rotate**	**3-Slot Compounder**	**Trading 212 API v0**
EMA 20/50, RSI, ATR	0-100 Quant Score	APPROVE / VETO / 1R	1.5% Risk & 5% DD Stop	Proxmox LXC & Web GUI

2. HIGH-BETA MOMENTUM UNIVERSE & 0-100 QUANT SCORING MODEL

Instead of slow defensive stocks or illiquid penny stocks with 2-5% spreads, the screener targets 25 liquid High-Beta Momentum leaders (3%-12% daily ATR range, <0.05% institutional bid-ask spread):

AI/Semis: PLTR, NVDA, AMD, SMCI, ARM, MU, AVGO, MRVL | Crypto Proxies: MSTR, COIN, MARA, RIOT
Space/Quantum: RKLB, IONQ, ASTS | Fintech/Cloud/EV: HOOD, SOFI, AFRM, UPST, SHOP, APP, TSLA, RIVN, CRWD, NET

QUANTITATIVE FACTOR	MAX PTS	SCORING & TECHNICAL CRITERIA
1. Trend & EMA Crossover	45 pts	45 pts: Fresh 20>50 EMA cross | 34 pts: Strong uptrend (Close > EMA20 > EMA50) | 20 pts: Coiling
2. RSI (14) Momentum	25 pts	25 pts: Sweet-spot RSI (54 to 67) | 18 pts: Positive RSI (50 to 54) | 0 pts: Overbought (RSI > 72)
3. Relative Volume (RVOL)	15 pts	15 pts: Volume >= 1.5x 20-bar average | 11 pts: Volume >= 1.1x average | 6 pts: Normal (>= 0.8x)
4. 10-Bar Rel. Strength	15 pts	15 pts: 10-bar Rate of Change (ROC) >= +1.5% | 11 pts: ROC >= +0.5% | 7 pts: Positive (> 0%)

RECOMMENDATION TIERS & CONCENTRATED 2-SLOT EXECUTION RULES:

STRONG BUY (>=80)	Auto-Traded (#1 Priority)	BUY (70-79)	Auto-Traded if slot open	WATCH	Wait for EMA cross

3. TRADE ENTRY, ATR STOP-LOSS, TAKE-PROFIT & 1R TRAILING STOP

A. Entry Conditions (Long-Only ISA/Invest)

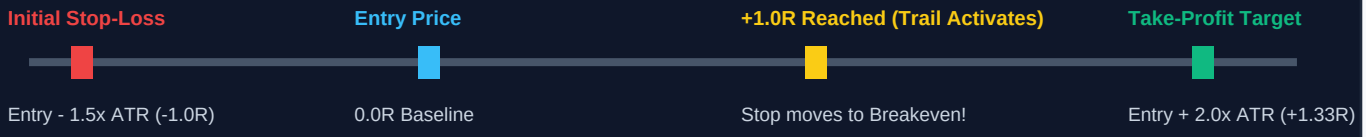
- Trigger 1 (Crossover): 20 EMA crosses above 50 EMA on a confirmed closed 15m bar AND 14-period RSI > 50.
- Trigger 2 (Top Quant Pick): Stock scores >= 70/100 with Close > 20 EMA > 50 EMA, rising 20 EMA, and RSI 50-70.
- Stocks with EMA20 <= EMA50 are held in WATCH status.

B. Dynamic Volatility Exit Rules (14-Period ATR)

- Initial Stop-Loss: Placed at Entry Price - (1.5 x ATR). Adapts automatically to each stock's volatility.
- Profit Target: Placed at Entry Price + (2.0 x ATR), providing a positive 1.33:1 reward-to-risk ratio.
- Slot Rotation: Freed cash rotates into next #1 pick.

HOW THE +1.0R RATCHETING TRAILING STOP WORKS (INTRA-BAR EVERY 15 SECONDS)

1R equals the initial risk distance: 1R = |Entry Price - Initial Stop-Loss| = 1.5 x ATR. Once unrealized profit reaches +1.0R, the bot activates Trailing Mode and ratchets the Stop-Loss upward at (Highest Price Reached - 1.5 x ATR), locking in gains:



4. COMPOUNDING POSITION SIZING & CAPITAL PRESERVATION GUARDS

1. 3-Slot Concentrated Compounding (£500 Base)

Concentrates capital into the top 3 setups (33% cap/slot):

```
Qty = min(Equity x 1.5% / StopDist, Equity x 33% / Price)
```

Every winning trade increases Total Equity, automatically compounding subsequent trade sizes without manual intervention.

2. Hard 5% 24-Hour Drawdown Circuit Breaker

RiskManager tracks a rolling 24-hour equity high-water mark.

```
Drawdown = (Peak_24h - Current_Equity) / Peak_24h
```

If drawdown reaches >= 5.0%, the bot immediately flattens all open positions via Market Sell and halts trading.

3. Slippage & Bid-Ask Spread Guard (0.15%)

Before any order is routed, top-of-book quotes are checked:

```
Spread % = (Ask - Bid) / Mid_Price <= 0.15%
```

Off-exchange IEX stub quotes are filtered out, and any instrument exceeding 0.15% spread is rejected.

4. Off-Hours Order Sync & T212 Rate Throttling

Reconciles both active working orders and open positions:

```
Sync = /equity/portfolio + /equity/orders
```

Preserves overnight queued orders until market open and enforces per-endpoint rate limits (5.2s cash, 50/min orders).

5. LIVE HIGH-BETA SCREENER & EXECUTION LIFECYCLE SNAPSHOT

Live examples from logs/trading_bot.log demonstrating high-beta selection, WATCH filtering, and stop-loss rotation:

TICKER	RATING	SCORE	PRICE	STOP-LOSS	TARGET	BOT ACTION & QUANTITATIVE THESIS
MU_US_EQ	STRONG BUY	88.0	$908.87	$901.76	$918.35	Auto-Bought Slot | Fresh 20/50 EMA cross + RVOL 1.24x
TSLA_US_EQ	STRONG BUY	81.0	$368.09	$365.55	$371.48	Auto-Bought #1 Pick | Uptrend + RSI 59.3 + RVOL 2.61x
APP_US_EQ	WATCH	71.0	$268.98	$267.13	$271.45	Held in WATCH | High RVOL 3.85x, waiting for EMA cross
PLTR_US_EQ	HOLD	55.0	$194.16	$192.59	$196.26	Skipped (<70) | Bullish EMA structure, RSI 51.8 neutral
WMT_US_EQ	SL EXIT	-1.0R	$118.67	$118.67	$119.63	Auto-Closed on intra-bar SL -> Rotated slot into MU_US_EQ

6. 24/7 PROXMOX LXC CONTAINER DEPLOYMENT & LAN WEB DASHBOARD

Always-On Headless Linux LXC Operation + Real-Time Web GUI (Port 8050)

Designed to run 24/7 inside an unprivileged Debian/Ubuntu LXC container on Proxmox VE (~110 MB RAM, 1 vCPU):

1. Systemd Daemon (t212-bot.service): Auto-starts on Proxmox boot and auto-restarts within 10s if interrupted.
2. LAN-Accessible Web Dashboard (0.0.0.0:8050): View Equity, Open Slots (x/2), 1R Trailing Gauges & 25-Stock Leaderboard.
3. One-Command Mac-to-Proxmox Deploy: push_to_proxmox.sh bundles code + .env and provisions the LXC via pct push/exec.
4. Live Trading FX Optimization: Hold USD in Trading 212 Invest when going live to avoid the 0.15% GBP/USD FX conversion fee.

```
./push_to_proxmox.sh root@<PROXMOX_IP> <CT_ID>  # 1-command deploy from Mac to Proxmox LXC container
systemctl status t212-bot                       # Check 24/7 systemd service inside Proxmox LXC
python main.py                                  # Run bot + LAN Web Dashboard (http://<IP>:8050)
python main.py --recommend                      # Print ranked 25-stock High-Beta leaderboard only
```